Robotics and Communications Systems Technology
RCET 2251: Systems Analog & Digital Theory

Day One Test Prep

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1. State Ohm's Law.

   Ohm's Law states that voltage is equal to current multiplied by resistance ($V = IR$).

2. State the electrical power equation.

   Electrical power is equal to voltage multiplied by current ($P = VI$).

3. State Kirchhoff's Voltage Law (KVL).

   The algebraic sum of all voltages around a closed loop is zero.

4. State Kirchhoff's Current Law (KCL).

   The algebraic sum of all currents entering and leaving a node is zero.

5. State Thevenin's Theorem.

   Any linear two-terminal circuit can be replaced by an equivalent circuit consisting of a single voltage source in series with an equivalent resistance.

6. State Norton's Theorem.

   Any linear two-terminal circuit can be replaced by an equivalent circuit consisting of a single current source in parallel with an equivalent resistance.

7. See Figure 1. Resistor Circuit, If $V_{DC} = 50V$, $R1 = 1K\Omega$, $R2 = 2K\Omega$, $R3 = 3K\Omega$, and the current through the load, $I_{RL} = 20mA$. Find the total circuit current I_{Total}.

   $I_{Total} = 25mA$

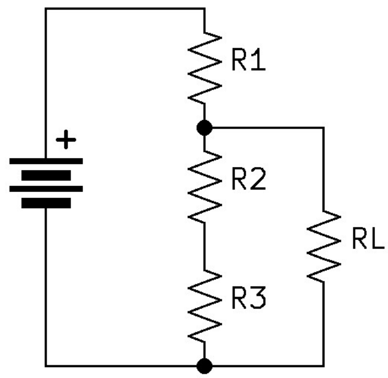

Figure 1: Resistor Circuit

8. See Figure 2. Resistor Circuit Analysis, If $VDC = 6V$, $R1 = 1K\Omega$, $R2 = 4K\Omega$, and $R3 = 4K\Omega$, what is V_{R3}

   $V_{R3} = 4V$

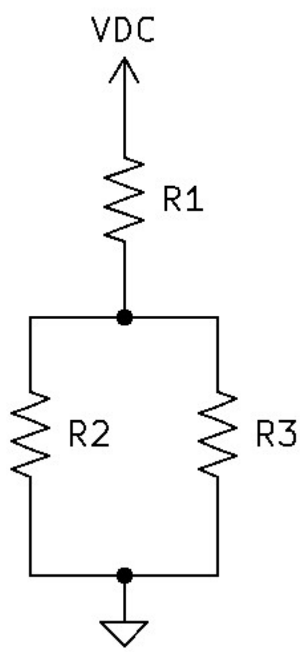

Figure 2: Resistor Circuit Analysis

9. See Figure 3. Resistor Circuit Analysis, If $VDC = 12V$, $R1 = 1.2K\Omega$, $R2 = 4.7K\Omega$, $R3 = 2K\Omega$, and $R4 = 2.7K\Omega$ Find: R_{Thev} and V_{Thev}.

   $R_{Thev} = 2.46K\Omega$ and $V_{Thev} = 3.1V$

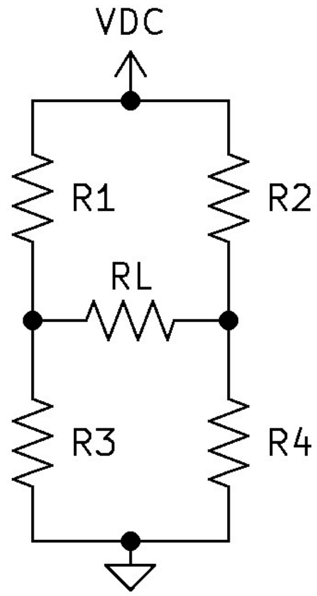

Figure 3: Resistor Circuit Analysis

10. See Figure 4. ABCD Circuit 1, If $VDC1 = 12V$, $VDC2 = 14V$, $R1 = 20\Omega$, $R2 = 40\Omega$, and $R3 = 60\Omega$
    Find: V_{AB}, V_{BC}, V_{CD}, V_{AD}
    Find: V_{BA}, V_{CB}, V_{DC}, V_{DA}
    Find: V_A, V_B, V_C, V_D

    $V_{AB} = 4.333V$, $V_{BC} = 8.666V$, $V_{CD} = 13V$, $V_{AD} = 26V$

    $V_{BA} = -4.333V$, $V_{CB} = -8.666V$, $V_{DC} = -13V$, $V_{DA} = -26V$

    $V_A = 12V$, $V_B = 7.667V$, $V_C = -1V$, $V_D = -14V$

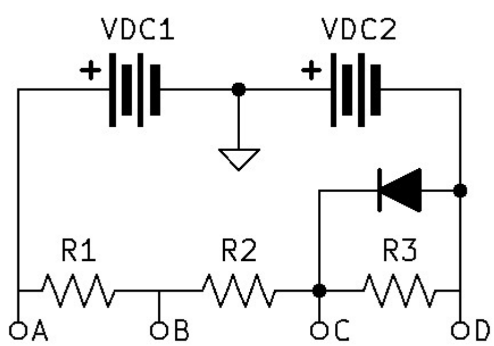

Figure 4: ABCD Circuit 1

11. See Figure 5. ABCD Circuit 2, If $VDC1 = 12V$, $VDC2 = 14V$, $R1 = 20\Omega$, $R2 = 40\Omega$, and $R3 = 60\Omega$
    Find: V_{AB}, V_{BC}, V_{CD}, V_{AD}
    Find: V_{BA}, V_{CB}, V_{DC}, V_{DA}
    Find: V_A, V_B, V_C, V_D

    $V_{AB} = 8.433V$, $V_{BC} = 16.867V$, $V_{CD} = 0.7V$, $V_{AD} = 26V$

    $V_{BA} = -8.433V$, $V_{CB} = -16.867V$, $V_{DC} = -0.7V$, $V_{DA} = -26V$

    $V_A = 12V$, $V_B = 3.567V$, $V_C = -13.3V$, $V_D = -14V$

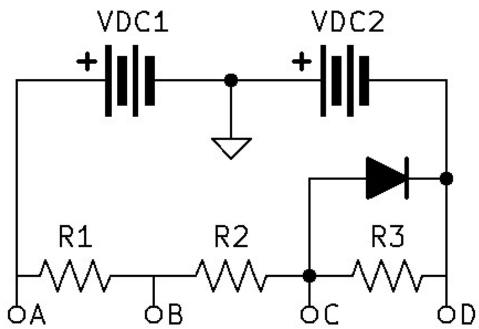

Figure 5: ABCD Circuit 2

12. Write the formula used to calculate capacitive reactance.

    $X_C = \frac{1}{2\pi FC}$

13. Write the formula used to calculate inductive reactance.

    $X_L = 2\pi FL$

14. Write the formula used to calculate frequency resonance in an RCL circuit.

    $F_R = \frac{1}{2\pi\sqrt{LC}}$

15. State the definition of impedance.

    Impedance is the total opposition to alternating current in a circuit and consists of both resistance and reactance, measured in ohms (Ω).

16. See Figure 6. Filter Identification 1, Identify the following frequency response (High-Pass or Low-Pass) for each circuit.

    A. = High-Pass
    B. = Low-Pass
    C. = Low-Pass
    D. = High-Pass

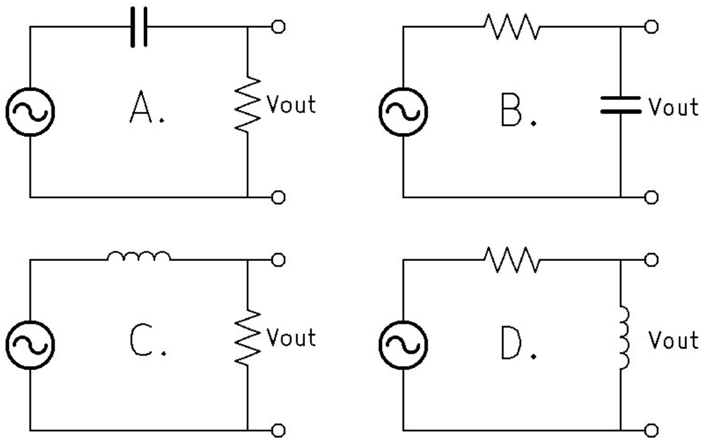

Figure 6: Filter Identification 1

17. See Figure 7. Filter Identification 2, Identify the following frequency response (Band-Pass or Band-Stop) for each circuit.

    A. = Band-Pass
    B. = Band-Stop
    C. = Band-Stop
    D. = Band-Pass

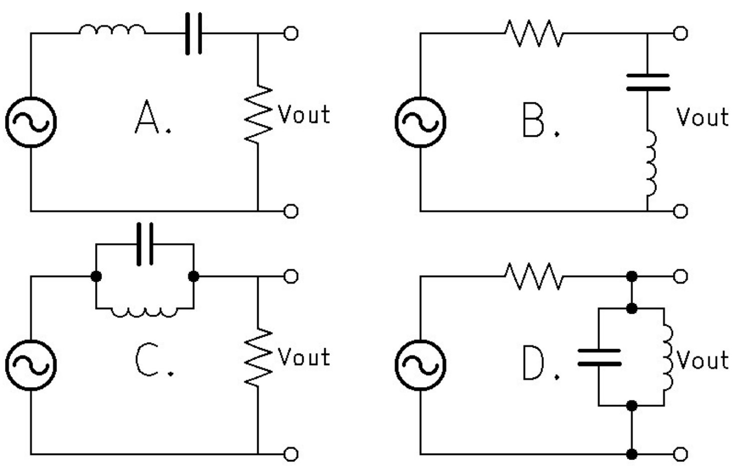

Figure 7: Filter Identification 2

18. See Figure 8. RC Circuit, If $V_{IN} = 5vp$, $C = 0.047\mu F$, and $R = 4.7K\Omega$, Find the critical frequency, the resistor, and the capacitor voltages at the critical frequency.

    $F_C = 720.5hz$
    $V_R = 3.535vp, \angle 45°$
    $V_C = 3.535vp, \angle - 45°$

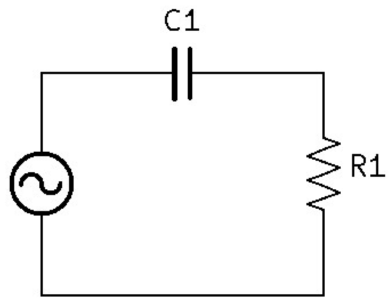

Figure 8: RC Circuit

19. See Figure 9. RCL Circuit, If $R1 = 1K\Omega$, $XC1 = 3.5K\Omega$, $R2 = 10K\Omega$, $XL1 = 4K\Omega$, $R3 = 20K\Omega$, and $XC2 = 10K\Omega$, find the equivalent series circuit resistance in Polar and Rectangular form.

    $Z_{total}(Polar) = 9.196K\Omega\angle - 16.437°$
    $Z_{total}(Rec) = 8.82K\Omega, -J2.603K\Omega$

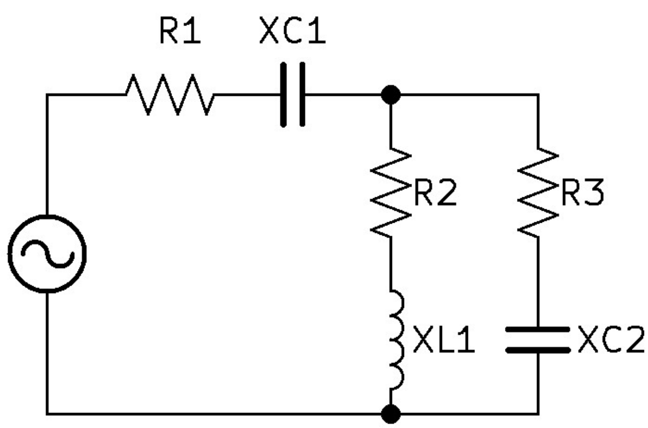

Figure 9: RCL Circuit

20. See Figure 10. RCL Circuit, If $V_{IN} = 10vp$, $Freq_{Gen} = 1Khz$, $R1 = 4.3K\Omega$, $R2 = 10K\Omega$, $R3 = 4.7K\Omega$, $C1 = 20nF$, $C2 = 8nF$, and $L = 3.5H$, find V_{R1}, V_{R2}, and V_{R3}, include the phase angle in reference to V_{Gen}.

    $V_{R1} = 1.064vp\angle 27.949°$
    $V_{R2} = 3.407vp\angle - 56.876°$
    $V_{R3} = 1.892vp\angle 85.378°$

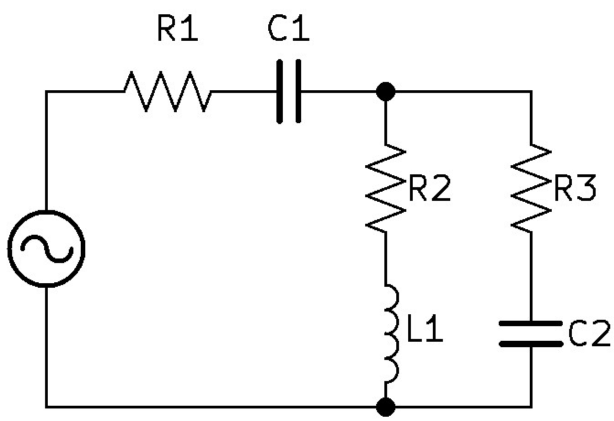

Figure 10: RCL Circuit

21. Write the equation for the capacitor charge formula.

    $V_C = V_{fin} - (V_{fin} - V_{int})e^{\frac{-t}{\tau}}$

22. See Figure 11. Capacitor Charge Circuit, If $V_{DC} = 10V$, $R1 = 1K\Omega$, and $C1 = 1\mu F$, Find the capacitor and resistor voltages at the following times: $500\mu Sec$, $1mSec$, $1.5mSec$, $2mSec$, $2.5mSec$, and $3mSec$.

    At $500\mu Sec$, $V_C = 3.935V$, $V_R = 6.065V$
    At $1.0mSec$, $V_C = 6.321V$, $V_R = 3.679V$
    At $1.5mSec$, $V_C = 7.769V$, $V_R = 2.231V$
    At $2.0mSec$, $V_C = 8.647V$, $V_R = 1.353V$
    At $2.5mSec$, $V_C = 9.179V$, $V_R = 0.821V$
    At $3.0mSec$, $V_C = 9.502V$, $V_R = 0.498V$

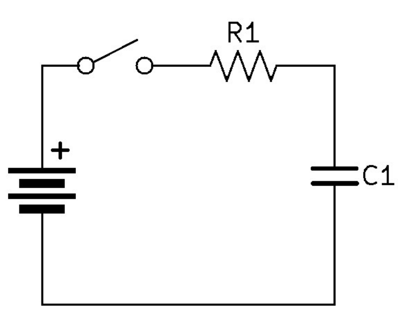

Figure 11: Capacitor Charge Circuit

23. See Figure 12. Pi-Filtered Rectifier, If $AC = 120V_{RMS}$, the turns ratio of the transformer is 4:1, $C1 = 22\mu F$, $R1 = 560\Omega$, $C2 = 47\mu F$, and $RL = 2.7K\Omega$, Find: $Vmax_A$, $Vmin_A$, VDC_A, $Vrpp_A$, VDC_B, $Vrpp_B$, and the output ripple factor ($r_{\%}$).

    $Vmax_A = 41.03vp$
    $Vmin_A = 36.53vp$
    $VDC = 38.78V$
    $Vrpp = 4.5Vpp$
    $VDC_B = 32.12V$
    $Vrpp_B = 225mVpp$
    $r_{\%} = 0.202\%$

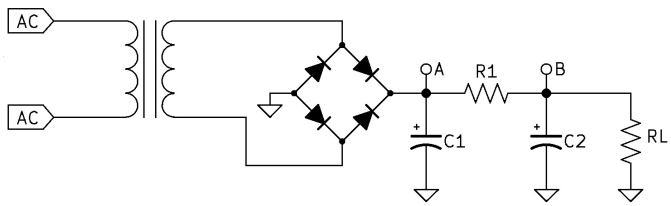

Figure 12: Pi-Filtered Rectifier

24. A full-wave rectifier has a $200\mu f$ capacitor and produces a ripple voltage that varies from 18V to 16.5V. If a 120VAC 60Hz source supplies the input, what is the value of the load resistor (RL)?

    $RL = 478.86\Omega$

25. See Figure 13. Bipolar Transistor Biasing Configurations, Identify the following Bipolar Transistor Biasing Configurations.

    A. = Base Bias
    B. = Universal Bias Or Voltage Divider Bias
    C. = Collector Feedback Bias
    D. = Emmitter Bias Or Two Power Supply Bias
    E. = Emmitter Bias Or Two Power Supply Bias

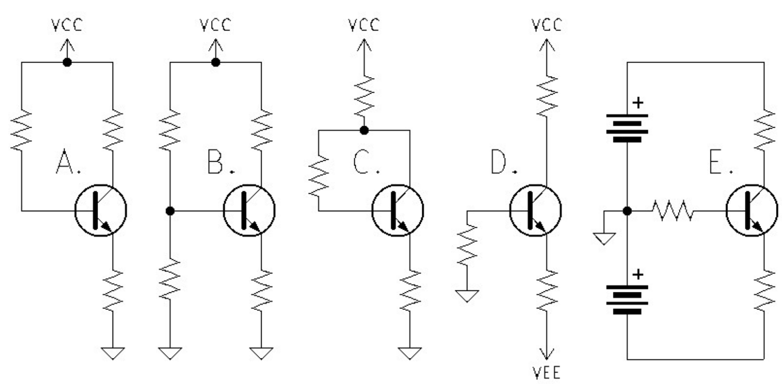

Figure 13: Bipolar Transistor Biasing Configurations

26. List Three Bipolar Junction Transistor Amplifier Configurations and their Characteristics.

    Common Emitter: Input on the Base, Output on the Collector, Good Voltage Gain (ΔV), Good Current Gain (ΔI), Great Power Gain (ΔP), High input impedance (Z_{IN}), and High output impedance (Z_{OUT}), the output waveform is 180° out of phase with the input.

    Common Collector: Input on the Base, Output on the Emitter, less than unity Voltage Gain ($\Delta V \approx 1$), Good Current Gain (ΔI), Good Power Gain (ΔP), High input impedance (Z_{IN}), and Low output impedance (Z_{OUT}), the output waveform is in phase with the input.

    Common Base: Input on the Emitter, Output on the Collector, Good Voltage Gain (ΔV), less than unity Current Gain ($\Delta I \approx 1$), Good Power Gain (ΔP), Low input impedance (Z_{IN}), and High output impedance (Z_{OUT}), the output waveform is in phase with the input.

27. See Figure 14. BJT Amplifier, If $VCC = 18V$, $R1 = 910K\Omega$, $R2 = 3K\Omega$, $R3 = 200\Omega$, $R4 = 800\Omega$, $RL = 3K\Omega$, and $\beta = 125$. At mid-band find: V_B, V_C, V_E, Z_{IN}, Z_{OUT}, ΔV, ΔI, ΔP,

    $V_B = 2.804V$, $V_C = 11.739V$, $V_E = 2.104V$
    $Z_{iN} = 25.993K\Omega$, $Z_{OUT} = 3K\Omega$
    $\Delta V = -7.005$, $\Delta I = -121.387$, $\Delta P = 850.316$

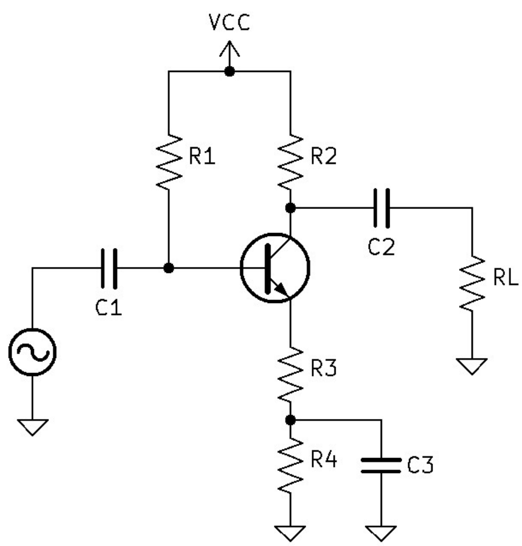

Figure 14: BJT Amplifier

28. See Figure 15. BJT Amplifier, If $VCC = 15V$, $Rgen = 50\Omega$, $R1 = 90K\Omega$, $R2 = 330\Omega$, $R3 = 120\Omega$, $RL = 120\Omega$, and $\beta = 100$. At mid-band find: Z_{IN}, Z_{OUT}, and gains,

    $Z_{iN} = 5.84K\Omega$, $Z_{OUT} = 2.29\Omega$
    $\Delta V = 0.970$, $\Delta I = 47.207$, $\Delta P = 45.79$

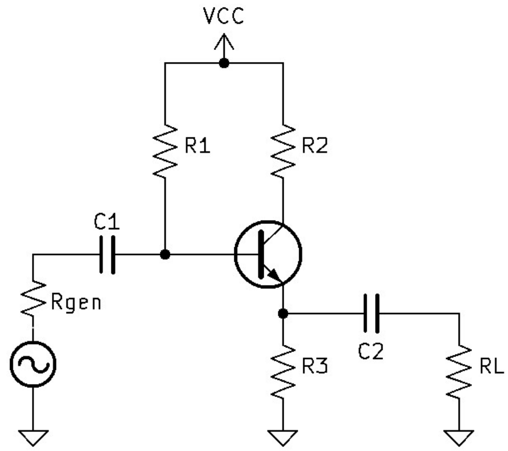

Figure 15: BJT Amplifier

29. See Figure 16. BJT Amplifier, If $VCC = 18V$, $Rgen = 75\Omega$, $R1 = 33K\Omega$, $R2 = 5.6K\Omega$, $R3 = 1.5K\Omega$, $R4 = 100\Omega$, $R5 = 290\Omega$, $RL = 1.5K\Omega$, and $\beta = 200$. At mid-band find: Z_{IN}, Z_{OUT}, and gains,

    $Z_{iN} = 77.427\Omega$, $Z_{OUT} = 1.5K\Omega$
    $\Delta V = 7.065$, $\Delta I = 0.365$, $\Delta P = 2.555$

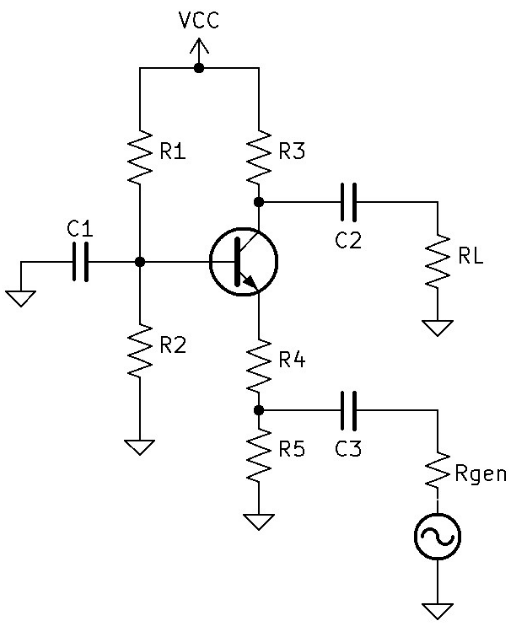

Figure 16: BJT Amplifier

30. List the four Characteristics of an Ideal Operational Amplifier.

    ∞ *Gain*
    ∞ *input impedance* Z_{IN}
    ∞ *Frequency Response – Bandwidth*
    0 *output impedance* Z_{OUT}

31. List the two rules for calculating Operational Amplifiers.

    No Signal Current into or out of the inputs.
    The output will do whatever it can to make the two input voltages the same.

32. List the Six Steps for calculating Operational Amplifiers.

    1. Find V_+
    2. w/feedback, $V_- = V_+$
    3. Find V_{RI}
    4. Find I_{RI}, $I_{RF} = I_{RI}$
    5. Find V_{RF}
    6. Kirchhoff to find V_{OUT}

33. See Figure 17. Operational Amplifier, If $V_{IN} = 1vp$, $RI = 1K\Omega$, and $RF = 10K\Omega$, find: ΔV and R_B.

    $\Delta V = -10$
    $R_B = 909\Omega$

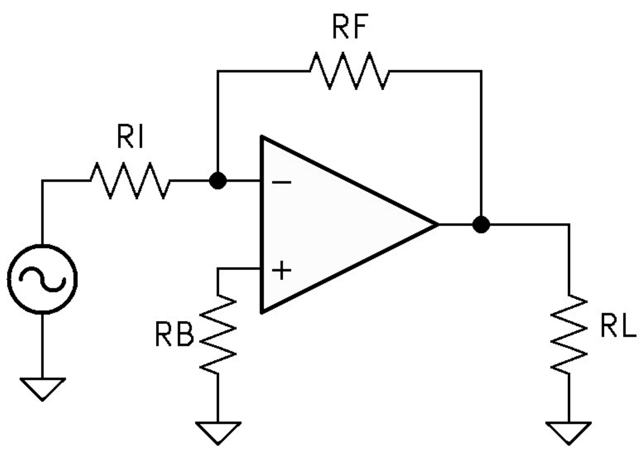

Figure 17: Operational Amplifier

34. See Figure 18. Operational Amplifier, If $V_{IN} = 1vp$, $RI = 1K\Omega$, and $RF = 10K\Omega$, find: ΔV and R_B.

$\Delta V = 11$
$R_B = 909\Omega$

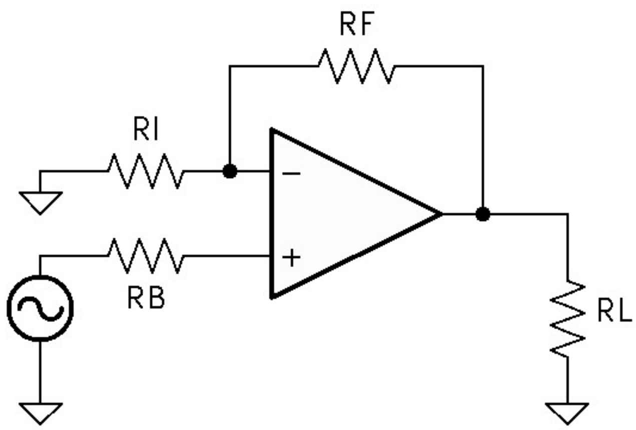

Figure 18: Operational Amplifier

35. See Figure 19. Operational Amplifier, If $V_1 = 900mV$, $V_2 = 600mV$, $V_3 = 300mV$, $R1 = 2K\Omega$, $R2 = 1.5K\Omega$, and $R3 = 1K\Omega$, find: V_{OUT} and R_B.

$V_{OUT} = -6.9V$
$R_B = 429\Omega$

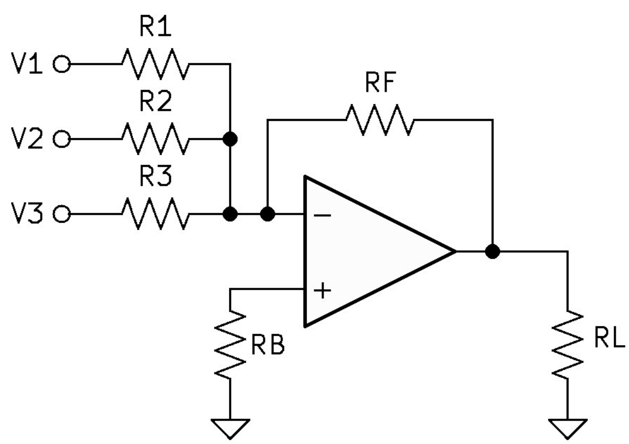

Figure 19: Operational Amplifier

36. See Figure 20. Operational Amplifier, If $V_1 = 4.5V$, $V_2 = 2V$, R1-R4 (all resistors) $= 1.5K\Omega$, find: V_{OUT}.

    $V_{OUT} = -2.5V$

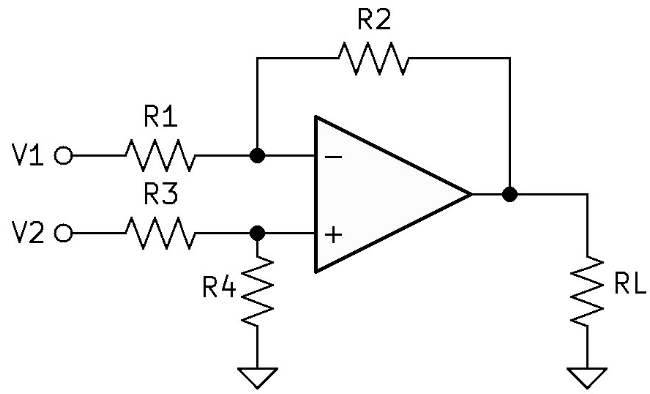

Figure 20: Operational Amplifier

37. See Figure 20. Operational Amplifier. See the previous question, find V_{OUT}, when the values of R2 and R4 are changed to $6K\Omega$. find: V_{OUT}.

    $V_{OUT} = -10V$

38. See Figure 21. Operational Amplifier, If $V_1 = 2V$, find: V_{OUT}.

    $V_{OUT} = 2V$

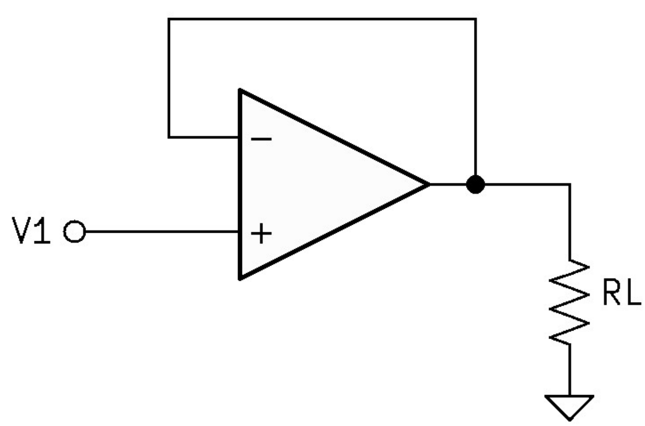

Figure 21: Operational Amplifier

39. Convert 37_{10} and 221_{10} to binary.

    $37_{10} = 100101_2$
    $221_{10} = 11011101_2$

40. Convert 63_8, $D9_{16}$, and 10011001_2 to decimal.

    $63_8 = 51_{10}$
    $D9_{16} = 217_{10}$
    $10011001_2 = 153_{10}$

41. List the following TTL values:$I_{OutH}Max$, $I_{OutL}Max$, $V_{OutH}Typ.$, and $V_{OutL}Typ.$

    $Iout_{HIGH}Max = 400\mu A$
    $Iout_{LOW}Max = 16mA$
    $Vout_{HIGH}Typ. = 3.4V$
    $Vout_{LOW}Typ. = 0.2V$

42. See Figure 22. Counter Circuit, If the clock frequency is 20KHz, find the counter's count sequence, modulus, and the MSB frequency.

    Down Counter, sequence is 6 to 0
    The Modulus is 7
    The MSB Frequency is 2.857KHZ

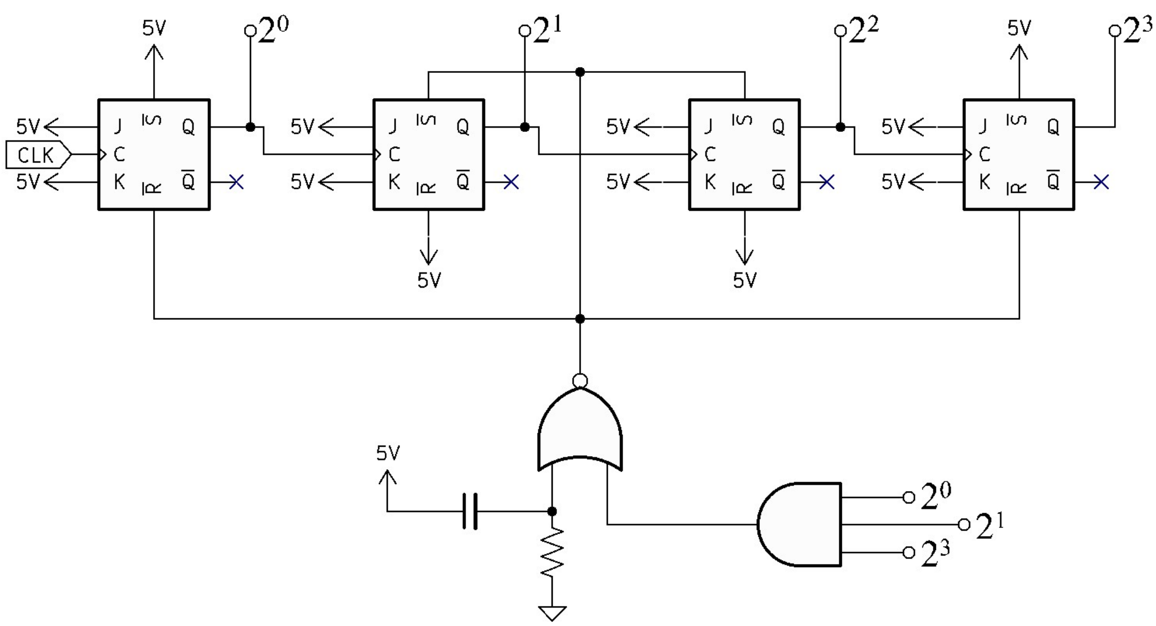

Figure 22: Counter Circuit

43. See Figure 23. Counter Circuit, If the clock frequency is 20KHz, find the counter's count sequence, modulus, and the MSB frequency.

Up Counter, sequence is 6 to 10
The Modulus is 5
The MSB Frequency is 4KHZ

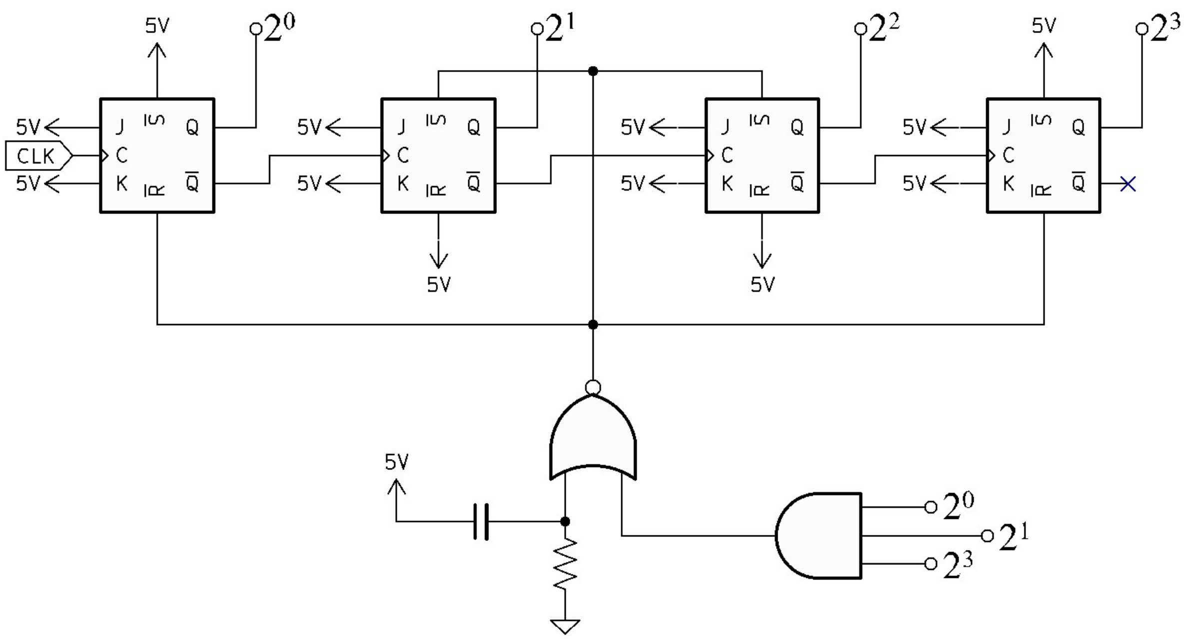

Figure 23: Counter Circuit